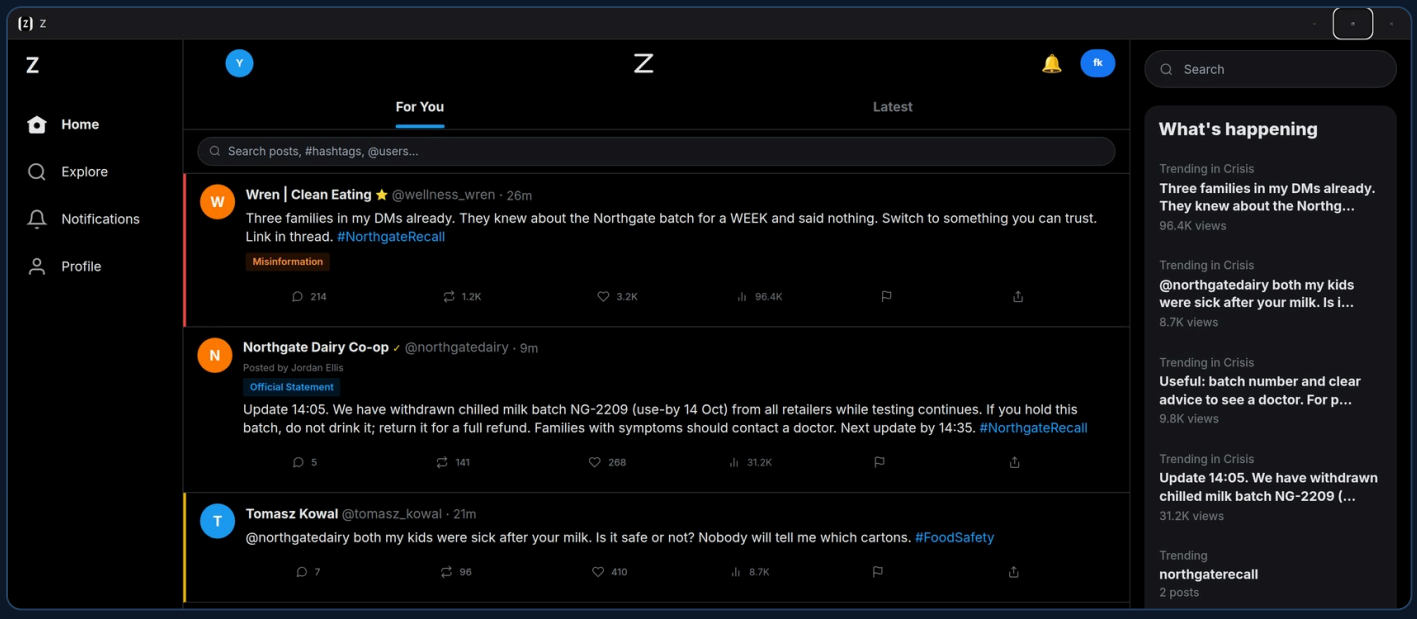
PROPHYION

AI-Driven Multi-Agent Crisis Simulation
Technical Case Study

Persistent AI Personalities • Agent Memory • Multi-Participant Interactions • Sentiment Tracking



The simulated social environment. Prophyion's own interface running with a fictional dairy-recall scenario and seeded demonstration data. Posts, replies and metrics shown here are authored fixtures; no live LLM call was made for this capture.

3 AI participant types: personas, stakeholders, teammates	**231** database migrations (Supabase / Postgres)	**283** automated tests, all passing	**30 s** simulation tick: injects, posts, sentiment

WHAT PROPHYION DOES

Prophyion is a browser-based crisis-communications training simulator. A team responds to a fictional incident while a simulated social environment (feed, DMs, email, team chat, news) is populated by AI characters. A trainer follows a live dashboard; an after-action review scores the response.

MY ROLE

CTO and technical lead, responsible for the platform architecture and technical delivery: NPC runtime, simulation state and sentiment logic, data model and trainer tooling.

VERIFIED TECHNOLOGY STACK

`React 19` `Vite 7` `TypeScript` `Tailwind` `Node.js + Express` `Socket.IO` `Zod` `Supabase: Postgres · Auth · RLS` `LLM client: OpenAI or AWS Bedrock`

The LLM client is provider-switchable with structured JSON output, an hourly budget and retries.

HOW HUMANS AND NPCS INTERACT

- **Humans act** by posting, replying, sending DMs and email, and using team chat.
- **Stakeholders** (a reporter, an official, a retailer) answer in character across email, team chat, messenger and phone, drawing on one shared conversation log per stakeholder.
- **Crowd personas** react to feed posts: they support, attack, cover or stay neutral, and a competitor persona positions itself.
- **AI teammates** can play alongside humans as optional, feature-flagged participants `EXPERIMENTAL`
- **The trainer** sees live sentiment, graded responses and inject state, and can steer the exercise.

Why this is multi-agent experience

Several LLM-driven characters, each with its own persona definition, share one simulated world. They read shared state (stored posts, conversation logs, session state), respond to the same events and obey the same server-side rules. Coordination is **server-orchestrated**: the platform decides who reacts and when. It is not free-form agent-to-agent dialogue, the main gap the proposed project would close (page 3).

What the repository actually implements

ARCHITECTURE: HUB-AND-SPOKE, SERVER-ORCHESTRATED

Triggers
Player post, reply, DM, email, chat
30 s scheduler tick
Scenario injects and their conditions

→

Orchestration
Pile-on reaction engine
Stakeholder reply service
Inject scheduler + conditions
pickResponders (seeded, capped, deterministic)

→

Agents (LLM-driven)
Crowd personas
Stakeholders with hidden character sheets
AI teammate bots (experimental, flagged)

→

Shared LLM client
OpenAI | Bedrock
JSON output, budget
Server-side guards
Verdict enforcement, scrubbing, rate caps

→

State (Postgres)
`social_posts`
`stakeholder_conversations`
`social_state`
`sentiment_snapshots`

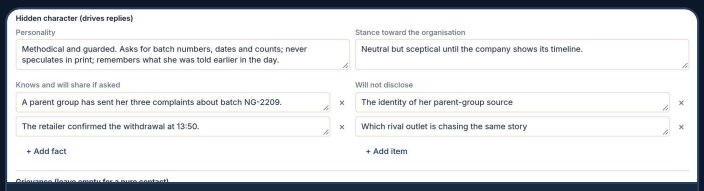
↓

Sentiment engine
Deterministic recompute every tick + LLM graders

←

Outputs
Trainer dashboard (Socket.IO) · after-action review

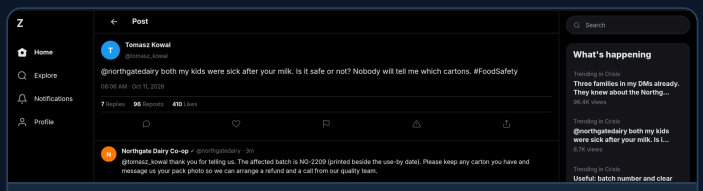
Threshold crossings generate consequence injects, which re-enter as triggers



Persistent persona definitions — IMPLEMENTED

Each stakeholder has a hidden character sheet: personality, stance, knowledge, what they will not disclose, grievance, resolution criteria, persuadability and hard constraints, validated by a Zod schema. Crowd personas add bias, backstory and posting pattern.

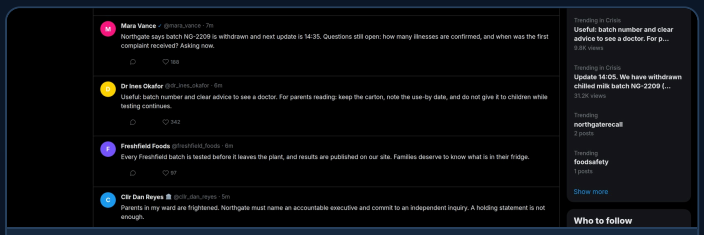
`Trainer scenario editor · fictional data`



Conversation memory — IMPLEMENTED · SESSION SCOPE

One log per stakeholder spans email, team chat, messenger and phone (`stakeholder_conversations`); recent turns feed every reply. Feed NPCs rebuild thread context from stored posts. Scope is one session: no cross-session memory.

`Thread continuity · authored replies`



Many NPCs, one event — IMPLEMENTED

One official statement triggers a pile-on. Per-persona gates (type, bias, format) choose who reacts, within a cap. One batched LLM call writes the replies, posted staggered over 3 to 30 s.

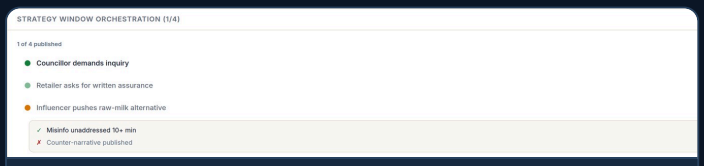
`Five personas reacting · authored replies`



Sentiment meters — IMPLEMENTED

Four session-level dimensions plus a weighted composite, recomputed deterministically each tick; two LLM graders feed in. A training metric, **not validated public-opinion measurement**.

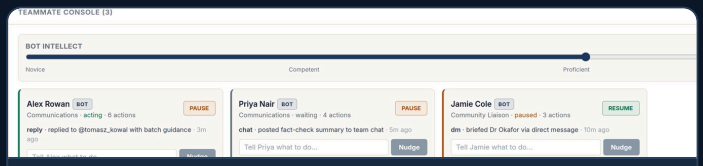
`Trainer dashboard · scripted series`



State-driven orchestration — IMPLEMENTED

A 30 s tick publishes injects whose conditions are met (sentiment thresholds, unaddressed misinformation, statements issued). The trainer panel shows each as published, partial or waiting.

`Strategy window orchestration · fictional data`



AI teammates (agent management) — EXPERIMENTAL

LLM teammates play alongside humans. An intellect slider (0 to 100) tunes behaviour, trainers can pause or nudge each bot, and an hourly LLM budget caps usage. Bot memory is in-process, rebuilt from the database on restart.

`Teammate console · fictional bots · feature-flagged`

Consistency and safety guards — IMPLEMENTED

Confirmed facts and a "do not contradict" rule in character prompts · server-side enforcement of LLM verdicts (persuadability, hard constraints, criteria) · placeholder scrubbing · reply coalescing and rate caps · an LLM watchdog that flags contradictions in a player's own statements.
Documented fix: NPCs wrote "Hi [Name]" because they did not know who was writing. The authenticated sender is now passed into the prompt and placeholders are scrubbed.

Not in Prophyion today

Cross-session memory · embedding retrieval · automated personality-drift scoring · autonomous agent-to-agent turn-taking (NPC exchanges are server-scheduled) · per-NPC sentiment · dollar-cost tracking (token usage is logged).

From Crisis Simulation Agents to Coached Competitive Agents

How the experience above maps to the requested negotiation platform. Column A is evidence from Prophyion; column B is new work that does not exist yet.

REQUIREMENT	A · DEMONSTRATED IN PROPHYION	B · NEW / PROPOSED
8–10 agents, separate personalities	IMPLEMENTED Structured persona definitions (personality, stance, knowledge, constraints) validated by schema; many personas active per event.	Versioned negotiation personas with explicit trait profiles and hidden priorities.
A human coach per agent; 5 stored lessons	PARTIAL Persona data persists in the database; trainers can nudge AI teammates (experimental). No coaching-lesson model exists.	Coach accounts, five lesson slots per agent, lesson edits written to a change history.
Persistent memory	IMPLEMENTED Cross-channel conversation log per stakeholder, session-scoped. AI teammates also track their own prior official statements.	Memory that persists across matches under explicit rules: what is stored, injected and forgotten.
2-agent duels with hidden priorities	PARTIAL Hidden "will not disclose", criteria and hard constraints, enforced server-side. One side is always a human.	Agent-vs-agent duel engine with private priority weights and a defined offer protocol.
5-agent group negotiations	PARTIAL Several NPCs respond to one event, capped and staggered. Orchestrated by the server, not turn-taking between agents.	Turn-taking design for five agents (two options below).
Deterministic scoring, independent of the LLM	IMPLEMENTED Sentiment composite and verdict enforcement are rule-based. Content grades from an LLM are a separate, labelled input.	Scores computed only from structured offers and rules; the LLM supplies language, never points.
Transcripts, memory changes, history	IMPLEMENTED Stored conversation logs, session events and an after-action review.	Immutable match transcripts plus a before/after diff of each agent's memory.
Coached vs uncoached comparison	Not in Prophyion.	Paired matches with identical seeds and scenarios, differing only in lessons.
API usage and match cost	PARTIAL Token usage logged per call site; hourly LLM budget cap. No dollar-cost ledger.	Per-match, per-agent usage and cost ledger with budget stops.

PROPOSED ARCHITECTURE PROPOSED

Agent Personality + Five-Lesson Memory
Trait profile, hidden priorities, five coach lessons, versioned.

▶

Conversation / Negotiation Engine
Duel and group protocols; LLM writes dialogue and structured offers.

▶

Deterministic Rules and Scoring
Validates offers, applies payoffs; no LLM in the score.

▶

Match History and Behavioral Analysis
Transcripts, memory diffs, cost, coached vs uncoached.

TWO APPROACHES FOR THE FIVE-AGENT GROUP PROPOSED

1 · Deterministic round-robin

A fixed speaking order rotates each round, so every agent gets a turn and every match is reproducible.

Strengths: simple to build and test, fair airtime, exact replay, easiest to score.

Trade-off: stiff; no interruptions or reactions in the moment.

Prophyion link: seeded, capped, deterministic responder selection.

2 · Intent-driven turn-taking with a floor manager

Each agent declares an intent (speak, propose, respond to X, pass) with an urgency. A rule-based floor manager grants the floor, with fairness caps on consecutive turns.

Strengths: more natural exchanges, agents can respond to each other.

Trade-off: more moving parts; replay needs seeded tie-breaks and a logged intent queue.

Suggested path: approach 1 as the scored baseline, approach 2 as an experimental mode.

My experience building Prophyion provides a practical foundation in persistent AI identities, multi-agent interactions, simulation state and measurable outcomes. The proposed negotiation prototype would be developed as a separate application with its own mechanics, memory rules and experimental controls.